Locally Private Causal Inference with Unknown Within-Block Interference: Design-Based Effects, DoI Models, and Minimax Limits

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Abstract

Local differential privacy protects each participant before their outcome reaches an analyst, whereas interference lets one participant's treatment change another's outcome. Combining the two is not automatic: with a fixed number of arbitrary interference blocks, direct and spillover effects cannot be estimated with uniformly shrinking uncertainty. We give a design-based solution for randomized experiments with independent interference blocks. Blocks are randomized to low or high treatment saturation, outcomes may depend arbitrarily on every assignment within a block, and only outcomes are privatized. A one-bit mechanism and Horvitz–Thompson scores identify saturation-specific direct effects and policy spillover effects without observing or correctly specifying the latent Degree of Interference (DoI). Cluster-score intervals account jointly for assignment dependence and privacy noise. For C blocks of m users, the worst-case mean squared error is, up to design-overlap constants, $\Theta\big(\min\{1, C^{-1} + (Cm\tanh^2(\epsilon/2))^{-1}\}\big)$. Thus interference limits independent replication while user-level privacy contributes a separate information loss; multiplying the two penalties would be incorrect. A second oracle lower bound shows where a multiplicative cost is real: an exposure cell of probability ρ requires risk $\Omega(\min\{1, (Cm\rho\tanh^2(\epsilon/2))^{-1}\})$. The risk grows exponentially until constant-risk saturation when a DoI cell specifies many peers' assignments. We also derive the exact privatized-bit likelihood for latent DoI outcome models. Simulations are consistent with the predicted scaling, show calibrated intervals in a fixed-population stress test, compare the one-bit mechanism with Laplace noise, and expose the bias–variance tradeoff of model-dependent outcome regression.

1 Introduction

Randomized experiments often collect sensitive outcomes in settings where people interact. A cash-transfer program may affect siblings, a vaccination may protect contacts, and an online intervention may propagate through a network. Local differential privacy (LDP) asks each user to randomize their own record before transmission, so the curator never receives the raw outcome. Interference breaks the usual assumption that a unit's outcome depends only on that unit's treatment. Each problem is well studied in isolation; together they create a basic question: what causal effect is still identified, and what information is lost to privacy versus interference?

Two recent frameworks make the question precise. Ohnishi and Awan [2025] derive locally private inference for randomized experiments under no interference. Separately, Ohnishi et al. [2025] introduce the Degree of Interference (DoI), a latent random function that compresses how other units' assignments affect one unit. Their DoI framework avoids declaring a possibly wrong exposure mapping, but its flexibility necessarily shifts some work to a model for outcomes given the latent DoI. A direct combination must therefore resolve three tensions. First, arbitrary interference can prevent concentration even with a large network [Basse and Airoldi, 2018]. Second, a latent representation cannot create identification from one observed assignment. Third, privacy noise and assignment dependence enter the risk through different channels.

We resolve these tensions by separating design-based claims from model-dependent sensitivity analysis. Our primary estimands average over randomized saturation policies. The experiment contains C independent blocks of m units. A block is randomized to a low or high treatment probability, then units are randomized within that block. Interference may be arbitrary and unknown inside each block, including nonlinear and higher-order effects. This design identifies two direct effects, one under each saturation policy, and two policy spillover effects, one for treated and one for untreated units. These quantities do not condition on a learned DoI value. A latent DoI model is optional: it can improve precision or describe assignment-conditional effects, but it is not used to justify the main causal identification claim.

The paper makes three claims.

1. **One private bit is enough for design-based inference.** For outcomes in $[0, 1]$, a tailored one-bit channel gives an unbiased pseudo-outcome. Plugging it into block-level Horvitz–Thompson scores [Horvitz and Thompson, 1952] yield exactly unbiased direct and policy-spillover estimators under arbitrary within-block interference. The sample variance of independent block scores gives a simple, privacy-aware standard error.
2. **Privacy and block interference create two different bottlenecks.** With $N = Cm$ users and privacy signal $\tanh(\epsilon/2)$, the matched minimax risk is $C^{-1} + [N \tanh^2(\epsilon/2)]^{-1}$, truncated at a constant. The first term counts independent interference blocks; the second counts independent local messages. A separate oracle bound for a DoI cell of probability ρ is $\min\{1, [N\rho \tanh^2(\epsilon/2)]^{-1}\}$. This identifies the genuine privacy–interference interaction: rare, highly specific exposures amplify privacy loss until risk reaches a constant.
3. **Latent DoI models have an exact LDP observation layer.** If an outcome model has conditional mean $\mu(z, g)$, the released bit is Bernoulli with probability $b_\epsilon + \tanh(\epsilon/2)\mu(z, g)$. This likelihood can wrap a dependent Dirichlet process or a finite-feature approximation without decoding negative pseudo-outcomes. We use a transparent finite-feature outcome regression to show both its possible point-precision gain and its misspecification risk. This regression is model-dependent and is not used for interval inference.

The scope is deliberately narrower than "arbitrary interference on one network." That goal cannot yield uniformly precise estimation without growing independent replication or structural assumptions. We require known interference-block boundaries and rule out cross-block interference; the interference function inside a block remains unknown. Treatments, block labels, network summaries, and design probabilities are public. Only each outcome receives an ϵ-LDP guarantee. These limits are part of the result, not implementation details.

2 Related work

Local privacy. Randomized response predates modern differential privacy [Warner, 1965]. Differential privacy formalized stability to changing one input [Dwork et al., 2006], and the local model was connected to statistical-query learning by Kasiviswanathan et al. [2011]. LDP minimax theory shows that privacy contracts the distinguishability of nearby distributions [Duchi et al., 2013, 2018]. Sharp mechanisms and contraction results appear in Kairouz et al. [2016b,a], Asoodeh and Zhang [2024]; broader geometric and instance-specific views are developed by Rohde and Steinberger [2020], Duchi and Ruan [2024]. Communication reductions and information contractions give complementary lower-bound tools [Duchi and Rogers, 2019, Acharya et al., 2020, Barnes et al., 2020, Raginsky, 2016]. Locally private mean intervals are studied by Gaboardi et al. [2019]. Ohnishi and Awan [2025] are the closest causal reference; their main setting invokes no interference, whereas our estimands, variance, and lower bounds operate on dependent blocks.

Interference and randomized designs. Potential-outcome definitions of direct and indirect effects date at least to Halloran and Struchiner [1995]. Partial interference and two-stage designs make such effects identifiable [Sobel, 2006, Hudgens and Halloran, 2008, Tchetgen Tchetgen and VanderWeele, 2012, Liu and Hudgens, 2014, Basse and Feller, 2018]. Randomized saturation and multilevel designs are studied by Sinclair et al. [2012], Baird et al. [2018]. Exposure restrictions give meaning to responses under interference [Manski, 2013], and general exposure-based design estimators are developed by Aronow and Samii [2017]. Graph-aware designs aim to increase useful exposure probabilities [Ugander et al., 2013, Eckles et al., 2017]. Randomization tests can probe interference without fully estimating an effect surface [Bowers et al., 2013, Athey et al., 2018].

Unknown or misspecified interference is especially relevant here. Unrestricted interference can defeat large-sample design-based intuition [Basse and Airoldi, 2018]. Sävje et al. [2021] characterize what conventional estimators mean under unknown interference, and Sävje [2024] separates effect definitions from exposure restrictions. Network asymptotics require explicit decay or graph conditions [Leung, 2020, 2022, Li and Wager, 2022]. Other work studies learned networks, network uncertainty, observational exposures, causal diagrams, treatment-allocation policies, and bipartite interference [Bhattacharya et al., 2020, Forastiere et al., 2021, Ogburn and VanderWeele, 2014, Papadogeorgou et al., 2019, Zigler and Papadogeorgou, 2021, Harshaw et al., 2023]. We instead obtain design-based validity by changing the experimental unit of replication from the person to the interference block.

Latent and Bayesian interference models. The DoI framework uses a dependent Dirichlet process mixture to represent an unknown interference channel [Ohnishi et al., 2025]. Related Bayesian analyses use generalized propensity scores or flexible models for two-stage experiments [Forastiere et al., 2022, Ohnishi and Sabbaghi, 2024]; blocked stick-breaking computation traces to Ishwaran and James [2001]. Our contribution is the LDP observation layer and a clear boundary between model-based assignment-conditional inference and design-based policy effects.

3 Setup: latent interference and policy effects

3.1 Randomized saturation design

There are C interference blocks, indexed by c, each containing $m \geq 2$ units indexed by i; $N = Cm$. Block c receives a public saturation arm $S_c \in \{0, 1\}$ with

$$S_c \sim \text{Bernoulli}(r), \qquad r \in (0, 1).$$

Conditional on $S_c = s$, treatments are assigned independently within the block,

$$Z_{ci} \mid S_c = s \sim \text{Bernoulli}(p_s), \qquad 0 < p_0 < p_1 < 1.$$

The two values p_0 and p_1 are the low and high saturation policies. Write $q_{s1} = p_s$, $q_{s0} = 1 - p_s$, $r_1 = r$, and $r_0 = 1 - r$.

For an assignment vector $\mathbf{z}_c \in \{0, 1\}^m$, unit (c, i) has potential outcome $Y_{ci}(\mathbf{z}_c) \in [0, 1]$. We make no restriction on how another assignment in the same block changes this outcome.

Assumption 1 (Independent blocks and randomized assignment)**.** Potential-outcome schedules and pretreatment characteristics are independent and identically distributed across blocks. There is no cross-block interference. The saturation and unit-level assignments are generated as above and are independent of all potential outcomes.

The identical-distribution condition is needed for the superpopulation central limit theorem and minimax statement. Exact unbiasedness conditional on a fixed collection of C schedules needs only independent randomization and no cross-block interference.

3.2 The Degree of Interference is latent

Let T_{ci} collect public pretreatment information, possibly including a network row. In the DoI framework, a random function $G_{ci}(T_{ci}, \mathbf{z}_{c,-i})$ compresses other units' assignments. We use the operational form

$$Y_{ci}(z, \mathbf{z}_{-i}) = \int \widetilde{Y}_{ci}(z, g)\, dF_{ci}(g \mid T_{ci}, \mathbf{z}_{-i}), \tag{1}$$

where F_{ci} is the latent DoI distribution [Ohnishi et al., 2025]. Equation (1) includes a deterministic exposure mapping as a point mass, but does not require one. Our design-based analysis allows F_{ci} and $\widetilde{Y}_{ci}$ to be arbitrary inside a block. The DoI is therefore a nuisance representation for the primary results, not an observed exposure.

3.3 Direct and policy-spillover estimands

For $s, z \in \{0, 1\}$, define the assignment-marginal cell mean

$$\mu_{sz} = \mathbb{E}\left[\frac{1}{m}\sum_{i=1}^{m} \mathbb{E}_{\mathbf{Z}_{c,-i}\sim \text{Bernoulli}(p_s)^{m-1}}\{Y_{ci}(z, \mathbf{Z}_{c,-i})\}\right]. \tag{2}$$

The outer expectation averages blocks in the target population. The direct effect under saturation s and the spillover effect of moving from low to high saturation while holding own treatment at z are

$$\tau_s^{\text{dir}} = \mu_{s1} - \mu_{s0}, \qquad \tau_z^{\text{sp}} = \mu_{1z} - \mu_{0z}. \tag{3}$$

These are policy effects: "spillover" means changing the distribution of peers' treatments, not conditioning on a particular latent DoI value. This distinction avoids exponentially rare exposure cells and gives the estimands meaning even if the latent interference model is wrong.

Proposition 2 (Why growing block replication is necessary)**.** *For each contrast in (3) under a fixed full-support design with $p_0 < p_1$, the class of bounded arbitrary within-block schedules contains a submodel whose minimax mean squared error is at least c/C, even when raw outcomes are observed. Hence keeping C fixed precludes uniformly consistent estimation with shrinking uncertainty.*

The block-shock submodel used in the proof of theorem 7 establishes the claim. This is a precision statement, not nonidentification: the Horvitz–Thompson estimator remains design-unbiased even at $C = 1$, but its worst-case risk cannot vanish. Independent randomized blocks provide the replication needed for concentration. Other routes are possible, including structural network decay, but must be stated explicitly [Basse and Airoldi, 2018, Leung, 2022].

4 Outcome-only local privacy

Definition 3 (Outcome-only local differential privacy)**.** A channel Q is ϵ-locally differentially private for the outcome if, for every $y, y' \in [0,1]$, public design record $d = (S, Z, T)$, and measurable output set A,

$$Q(A \mid y, d) \leq e^{\epsilon} Q(A \mid y', d).$$

The released messages are conditionally independent across users given the raw experiment.

This guarantee does not protect S, Z, T, the network, or block membership. It limits leakage from a participant's own message; correlated neighbors' messages can still reveal information about that participant, so this is not network or group privacy. Protecting those variables or inferential channels requires a different mechanism and generally a larger error.

Table 1: **Scope of the design-based claims.** Here public means available to the analyst without a privacy guarantee.

Component	Status	Consequence
Outcome $Y \in [0,1]$	Private	One noninteractive ϵ-LDP bit per participant
Assignment and design	Public	S, Z, r, p_0, p_1 enter known inverse-probability weights
Blocks and covariates	Public	Block boundaries, T, and any network summaries are not protected
Interference	Restricted only across blocks	Arbitrary unknown response to every assignment inside a block; no cross-block effects
Primary estimands	Design-marginal	Direct and spillover contrasts average over the two randomized saturation policies
Latent DoI model	Optional	Model-dependent; not required for identification, unbiasedness, or the cluster-score intervals

Define

$$a_\epsilon = \tanh(\epsilon/2), \qquad b_\epsilon = \frac{1}{1+e^{\epsilon}}.$$

Each user with outcome $Y \in [0,1]$ releases one bit

$$R \mid Y = y \sim \text{Bernoulli}(b_\epsilon + a_\epsilon y)\,. \tag{4}$$

The analyst decodes the bit as

$$U = \frac{R - b_\epsilon}{a_\epsilon}. \tag{5}$$

Lemma 4 (Privacy and unbiased decoding)**.** *The channel in* (4) *is ϵ-LDP, $\mathbb{E}(U \mid Y) = Y$, and $\text{Var}(U \mid Y) \leq (4a_\epsilon^2)^{-1}$.*

At $y = 0$ and $y = 1$, the likelihood ratio for either bit equals e^{ϵ} in one direction; every intermediate input is a convex combination. The decoder can be negative or larger than one, which is harmless: it is only a post-processed estimating variable. As $\epsilon \downarrow 0$, $a_\epsilon \sim \epsilon/2$, recovering the usual ϵ^{-2} private mean-estimation cost. As $\epsilon \to \infty$, $a_\epsilon \to 1$.

5 Design-based estimators and inference

For cell (s, z), define

$$\widehat{\mu}_{sz} = \frac{1}{Cm} \sum_{c=1}^{C} \sum_{i=1}^{m} \frac{\mathbf{1}\{S_c = s, Z_{ci} = z\}}{r_s q_{sz}} U_{ci}. \tag{6}$$

The four effect estimators replace each μ in (3) by $\widehat{\mu}$. The weights use only known randomization probabilities.

Theorem 5 (Exact unbiasedness under unknown interference)**.** *Under assumption 1 and* (4)*,* $\mathbb{E}(\widehat{\mu}_{sz}) = \mu_{sz}$*. Consequently, every direct and spillover estimator is unbiased for its target in* (3)*. No DoI value, exposure mapping, outcome model, or network is required.*

For a target contrast $\tau = \sum_{s,z} \ell_{sz}\mu_{sz}$, let the block score be

$$H_c = \frac{1}{m} \sum_{i=1}^{m} \sum_{s,z} \ell_{sz} \frac{\mathbf{1}\{S_c = s, Z_{ci} = z\}}{r_s q_{sz}} U_{ci}, \qquad \widehat{\tau} = C^{-1} \sum_{c} H_c. \tag{7}$$

Use the cluster-score variance estimator

$$\widehat{V}(\widehat{\tau}) = \frac{1}{C(C-1)} \sum_{c=1}^{C} (H_c - \widehat{\tau})^2. \tag{8}$$

It automatically includes both assignment covariance inside a block and independent privacy noise. It does not transplant an independence variance formula from a no-interference setting.

Theorem 6 (Block central limit theorem)**.** *Fix m, $\epsilon > 0$, and design probabilities in $(0, 1)$. If the block scores have nonzero variance, then as $C \to \infty$,*

$$\frac{\widehat{\tau} - \tau}{\sqrt{\widehat{V}(\widehat{\tau})}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Thus Wald intervals based on (8) *are asymptotically valid. In experiments we use the t_{C-1} rather than normal critical value.*

The theorem is a superpopulation result for iid blocks. Conditional on one heterogeneous finite collection of schedules, the score variance can include between-block mean heterogeneity and is generally conservative rather than equal to the finite-population randomization variance. Our coverage experiment below is therefore a conditional fixed-population stress test, not a direct proof of the theorem.

6 Minimax limits: two bottlenecks and an overlap penalty

Let $\eta, \delta > 0$, restrict $r, p_0, p_1 \in [\eta, 1-\eta]$, and require $p_1 - p_0 \geq \delta$. Let $\mathcal{P}_{C,m}$ contain all independent-block superpopulation laws satisfying assumption 1 with outcomes in $[0, 1]$. For any one contrast in (3), define the noninteractive outcome-LDP minimax risk

$$\mathcal{R}^*_{C,m}(\epsilon) = \inf_{Q \in \mathcal{Q}_\epsilon} \inf_{\widehat{\tau}} \sup_{P \in \mathcal{P}_{C,m}} \mathbb{E}_{P,Q}(\widehat{\tau} - \tau(P))^2.$$

Theorem 7 (Matched privacy–block-interference rate)**.** *For constants $0 < c_{\eta,\delta} < C_\eta < \infty$ depending only on the displayed design bounds,*

$$c_{\eta,\delta} \min\left\{1, \frac{1}{C} + \frac{1}{Cma_\epsilon^2}\right\} \le \mathcal{R}^*_{C,m}(\epsilon) \le C_\eta \min\left\{1, \frac{1}{C} + \frac{1}{Cma_\epsilon^2}\right\}. \tag{9}$$

The unprojected estimator in (6) is exactly unbiased. Clipping its final contrast to $[-1, 1]$ destroys exact unbiasedness but cannot increase squared error and attains the capped upper bound.

The rate has a useful interpretation. The C^{-1} term is an interference term: an adversarial schedule can make all m outcomes in a block share one latent shock, so even raw outcomes provide only C independent draws. The $(Cma_\epsilon^2)^{-1}$ term is a privacy term: conditional on the raw outcomes, Cm users still contribute independent privacy coins. These terms are additive, or equivalently their maximum up to a factor of two. A product $m/(N\epsilon^2)$ would describe one private message per block, not the user-level privacy model studied here.

The DoI creates a different penalty when the target conditions on a specific latent exposure. For fixed $0 < \rho \le 1$, let $\mathcal{E}_{N,\rho}$ be the class of iid pairs (E_i, Y_i) with public exposure labels, $\mathbb{P}(E_i = e) = \rho$, outcomes $Y_i \in [0, 1]$, and otherwise unrestricted conditional outcome laws. The local channel may depend on public E_i but protects Y_i. Let $\mu_e = \mathbb{E}(Y_i \mid E_i = e)$.

Theorem 8 (Privacy–exposure overlap lower bound)**.** *For the oracle-observed exposure problem just defined, the outcome-LDP minimax risk satisfies*

$$\inf_{Q \in \mathcal{Q}_\epsilon} \inf_{\widehat{\mu}_e} \sup_{P \in \mathcal{E}_{N,\rho}} \mathbb{E}_{P,Q}(\widehat{\mu}_e - \mu_e(P))^2 \ge c \min\left\{1, \frac{1}{N\rho a_\epsilon^2}\right\}. \tag{10}$$

The Horvitz–Thompson one-bit estimator has the matching rate. Since the bound grants the analyst the true exposure label, it also lower-bounds any procedure that must infer a latent DoI cell.

If e specifies one exact pattern among d independently randomized peers, then $\rho = p^k(1-p)^{d-k}$. At $p = 1/2$, $\rho = 2^{-d}$ and the lower bound grows as $\min\{1, 2^d/(Na_\epsilon^2)\}$. The exponential growth before constant-risk saturation is the genuine multiplicative privacy–interference cost. Policy means in (2) integrate over these assignments and avoid conditioning on an exponentially rare cell. The choice is scientific: exact-exposure effects may be required, but their information cost should be reported.

7 An exact LDP likelihood for latent DoI models

The design-based estimator answers policy questions without estimating G. A scientist may also want assignment-conditional surfaces or improved point precision. Let a latent DoI model specify

$$G_{ci} \mid T_{ci}, \mathbf{Z}_{c,-i} \sim F_\phi, \qquad \mathbb{E}(Y_{ci} \mid Z_{ci} = z, G_{ci} = g, T_{ci}) = m_\theta(z, g, T_{ci}) \in [0, 1].$$

Under the one-bit mechanism, augmentation by G_{ci} gives the exact observation model

$$R_{ci} \mid z, g, T_{ci}, \theta \sim \text{Bernoulli}(b_\epsilon + a_\epsilon m_\theta(z, g, T_{ci})). \tag{11}$$

Marginalizing g replaces m_θ by its expectation under F_ϕ. This layer can be inserted into the dependent Dirichlet process mixture of Ohnishi et al. [2025]. It uses the released bit directly, avoids treating the unbounded decoder U as a Gaussian response, and costs no additional privacy because all fitting and posterior prediction are post-processing. Equation (11) is a unit-level conditional observation layer, not a claim that privatized outcomes are marginally independent. A full latent model with shared block shocks needs a joint block likelihood; a product of unit contributions is otherwise a composite likelihood and requires a block sandwich for uncertainty.

For a transparent empirical sensitivity analysis, we fit a finite-feature outcome regression rather than the full latent DoI process. Its public dictionary $\varphi(T, \mathbf{Z}_{-i})$ contains one-hop and strict two-hop treated fractions, generic quadratic and cubic terms, a public pretreatment covariate, and own-treatment interactions. It deliberately excludes the true generating threshold and sinusoidal terms. The conditional mean is

$$m_\theta = \text{logit}^{-1}\{\theta^\top \varphi(T, \mathbf{Z})\}.$$

Penalized maximum likelihood under (11) is followed by g-computation over the two saturation policies. We compare the private fit with the same generic regression on raw outcomes to isolate privacy loss, and with a no-interference private regression containing only the public covariate and own treatment. This is a DoI-motivated observed-feature approximation, not a fit of the latent Dirichlet-process model. None of these plug-in estimates is design-unbiased, and we do not attach causal confidence intervals to them; the comparison is a point-risk sensitivity analysis.

8 Experiments

The experiments ask four questions: (i) Does the one-bit estimator have negligible Monte Carlo bias and near-nominal coverage under nonlinear unknown interference? (ii) How much does it improve on outcome-level Laplace noise? (iii) Do sample size, block size, privacy, and exact exposure probability behave consistently with the predicted rates? (iv) When does DoI-motivated outcome regression help or hurt point estimation? All reported points aggregate independent Monte Carlo replications; error bars on RMSE and MSE are 95% Monte Carlo intervals, and coverage bars are 95% Wilson intervals across replications. The code uses fixed versioned seeds and writes checksums for every result file.

8.1 Data-generating processes and baselines

The main nonlinear population has $C = 200$ blocks of $m = 8$ units. Each block is a ring with local edges and random chords. Let $e_{1,ci}$ and $e_{2,ci}$ be one-hop and strict two-hop treated fractions. The latent DoI and bounded potential outcome are

$$\begin{aligned} G_{ci} &= 0.55e_{1,ci} + 0.25e_{2,ci} + 0.20\mathbf{1}\{e_{1,ci} > 0.45\}, \\ Y_{ci}(\mathbf{z}_c) &= \text{logit}^{-1}\big[-1.15 + X_{ci} + 0.55z_{ci} + 1.10G_{ci} + 0.35z_{ci}(G_{ci} - 0.5) + 0.25\sin(2\pi e_{1,ci})\big], \end{aligned}$$

where the fixed public covariate X_{ci} was drawn once from $\mathcal{N}(0, 0.2^2)$. Exact enumeration of all within-block peer assignments gives the finite-population truth. The design uses $p_0 = 0.2$, $p_1 = 0.8$, and $r = 0.5$. We use 800 replications at $\epsilon \in \{0.25, 0.5, 1, 2, 4\}$.

The scaling population uses complete within-block interference and

$$Y_{ci}(\mathbf{z}_c) = 0.28 + X_{ci} + 0.12z_{ci} + 0.22\bar{z}_{c,-i} + 0.06z_{ci}\bar{z}_{c,-i}, \qquad X_{ci} \sim \text{Unif}(-0.04, 0.04),$$

so the policy effects are analytic. We vary C at fixed $m = 8$, then vary m at fixed $N = 1600$, with 400 replications per condition. The exposure-overlap experiment observes an exact all-treated peer cell with probability 2^{-d}, uses $N = 100{,}000$ Bernoulli outcomes of mean 0.6, and has 2,000 replications. It is an oracle missing-cell subproblem, not a network simulation: the exposure label is public and own treatment is fixed. The DoI-feature study uses 150 replications.

Baselines are deliberately matched to the privacy scope: (a) the same design estimator with raw outcomes; (b) sensitivity-one Laplace noise with the same outcome-only ϵ; and (c) the released bit without the required debiasing. For the model-based sensitivity study we compare the design estimator, the generic private-likelihood regression, the same generic regression on raw outcomes, and a private no-interference regression.

8.2 Results

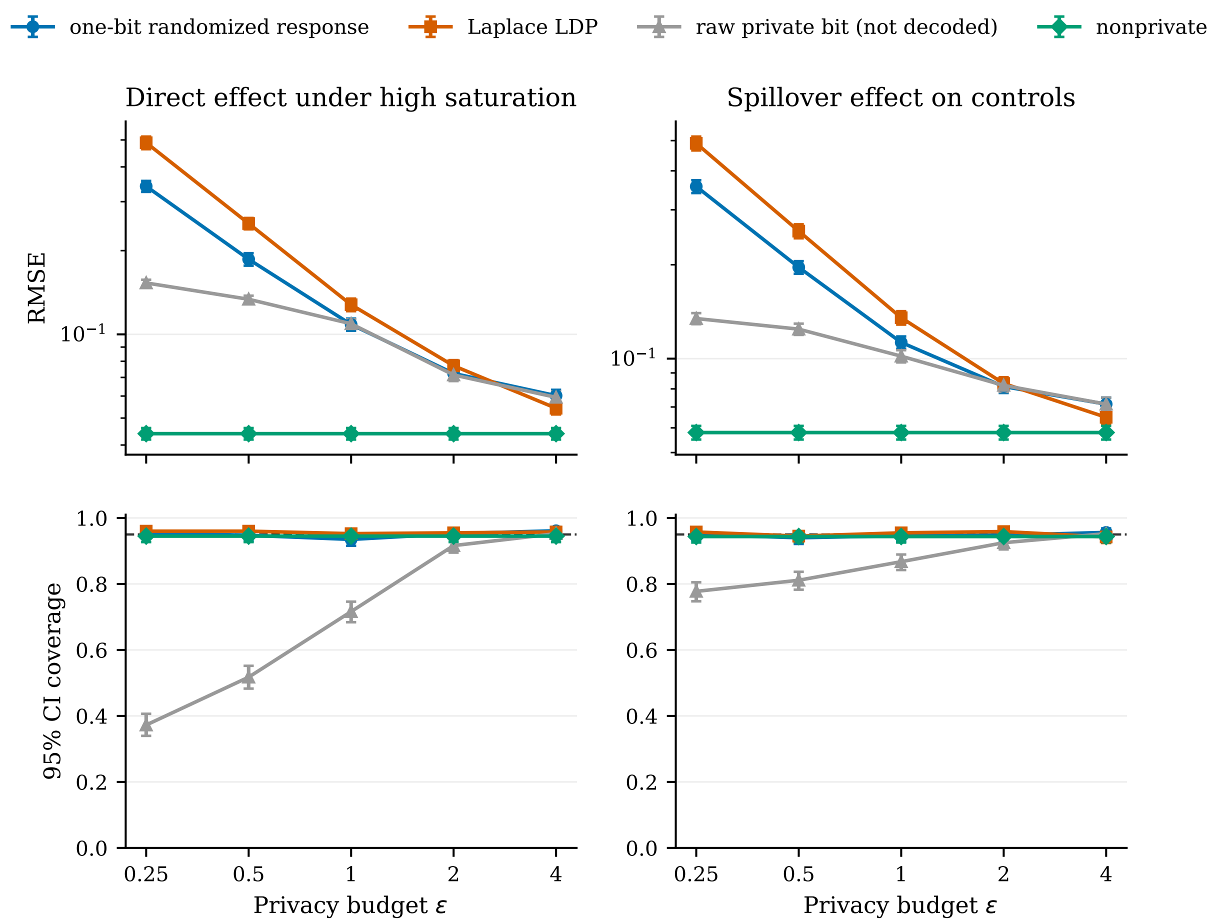

Figure 1: **Privacy–accuracy and coverage under nonlinear latent interference.** Points show unprojected Monte Carlo RMSE (top) and cluster-score interval coverage (bottom); bars are 95% Monte Carlo or Wilson intervals. The one-bit estimator is unbiased after decoding and is most advantageous over Laplace noise at tight privacy. The raw private bit, shown only as a negative control, is attenuated toward zero. The dashed line marks 95% coverage.

Figure 1 supports the first two empirical claims. Across both a direct and a spillover target, one-bit RMSE falls toward the nonprivate floor as ϵ grows. At tight privacy it is substantially smaller than Laplace RMSE, while the raw private bit can look low-variance only because it is biased toward

Table 2: Nonlinear benchmark diagnostics from 800 Monte Carlo replications. Projected RMSE clips only the point estimate to $[-1, 1]$; design-based intervals use unprojected cluster scores. SE/SD is the average estimated standard error divided by the empirical standard deviation.

Effect	ϵ	Method	Bias	RMSE	Projected RMSE	SE/SD	Coverage	Mean length
Direct, high saturation	–	Nonprivate	-0.000	0.044	0.044	0.97	0.945	0.169
	0.25	One-bit LDP	+0.007	0.341	0.341	1.04	0.955	1.405
	0.25	Laplace LDP	+0.016	0.489	0.466	1.02	0.960	1.975
	0.5	One-bit LDP	+0.014	0.186	0.186	0.99	0.948	0.729
	0.5	Laplace LDP	+0.003	0.250	0.250	1.01	0.960	0.995
	1	One-bit LDP	-0.004	0.109	0.109	0.96	0.935	0.411
	1	Laplace LDP	-0.001	0.128	0.128	1.03	0.953	0.520
Spillover on controls	–	Nonprivate	-0.002	0.058	0.058	0.99	0.944	0.227
	0.25	One-bit LDP	-0.019	0.356	0.355	1.01	0.950	1.415
	0.25	Laplace LDP	-0.026	0.490	0.469	1.03	0.958	1.980
	0.5	One-bit LDP	-0.010	0.196	0.196	0.96	0.940	0.744
	0.5	Laplace LDP	-0.010	0.256	0.256	0.99	0.945	1.006
	1	One-bit LDP	+0.001	0.113	0.113	0.98	0.946	0.438
	1	Laplace LDP	-0.002	0.135	0.135	1.02	0.955	0.543

zero. Cluster-score intervals maintain coverage near the nominal level in this fixed-population stress test; treating users as independent would ignore assignment covariance inside blocks.

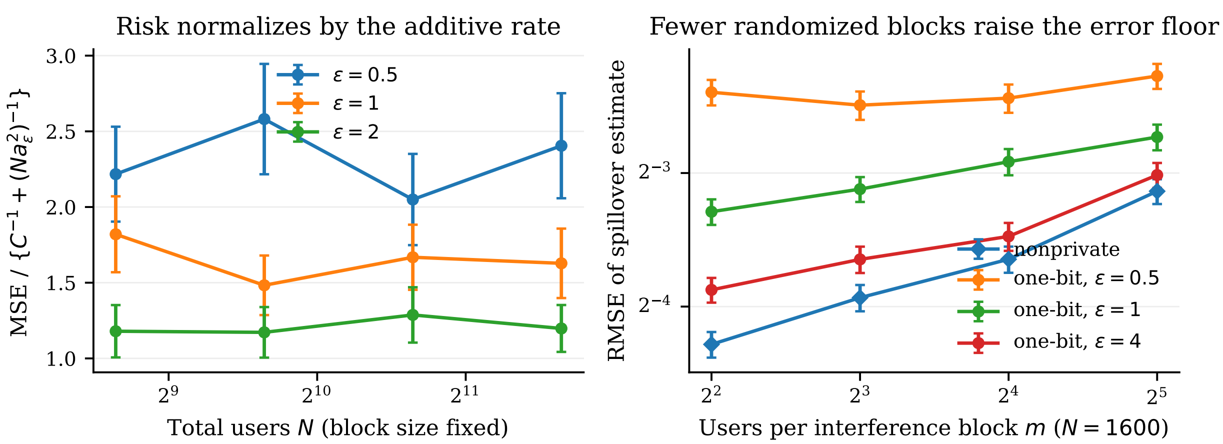

Figure 2: **Diagnostics for the two risk components.** Left: observed one-bit MSE divided by $C^{-1} + (Na_\epsilon^2)^{-1}$ stays within a budget-specific constant as N and C grow with $m = 8$. Right: at fixed N, increasing m reduces the number of first-stage randomized blocks and raises the nonprivate floor, while tight-privacy error is comparatively flat.

The normalized risk in the left panel of figure 2 is stable across a fourfold range of N, with no curve fitted to the simulated values. The right panel separates the two resources hidden by a conventional sample-size label: at fixed N, larger blocks mean fewer first-stage assignments and a larger error floor. This behavior is consistent with the upper-rate prediction; simulation does not prove the minimax lower bound.

Figure 3 illustrates the second lower-bound phenomenon. Every two additional peers quarter the expected cell count and approximately quadruple MSE. Local privacy shifts each curve upward by the expected a_ϵ^{-2} factor; it does not change the geometric dependence on exposure rarity before constant-risk saturation.

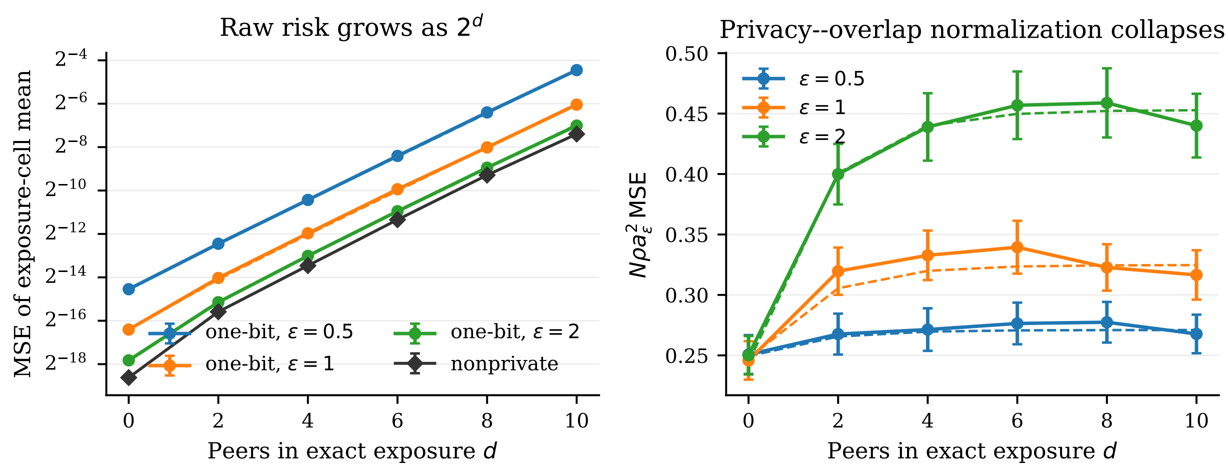

Figure 3: **Privacy multiplies exact-exposure rarity.** An exact assignment pattern among d peers has probability $\rho = 2^{-d}$. Left: HT mean squared error grows geometrically; dashed curves are exact finite-N moment calculations, not fits. Right: the normalized quantity $N\rho a_\epsilon^2$MSE follows those calculations and is nearly constant in d after the dense-cell edge case. Policy-marginal effects avoid selecting this rare cell.

The generic model in figure 4 approximates, but does not span, the generating threshold and sinusoid. It improves point RMSE in this population. The no-interference regression has smaller direct-effect variance at tight privacy only by accepting bias and estimates the spillover as zero, 0.110 below its truth. This is why plug-in fit is not our identification or interval argument. DoI-motivated models have a complementary role: describe and stabilize a design-identified policy effect, with feature sensitivity reported.

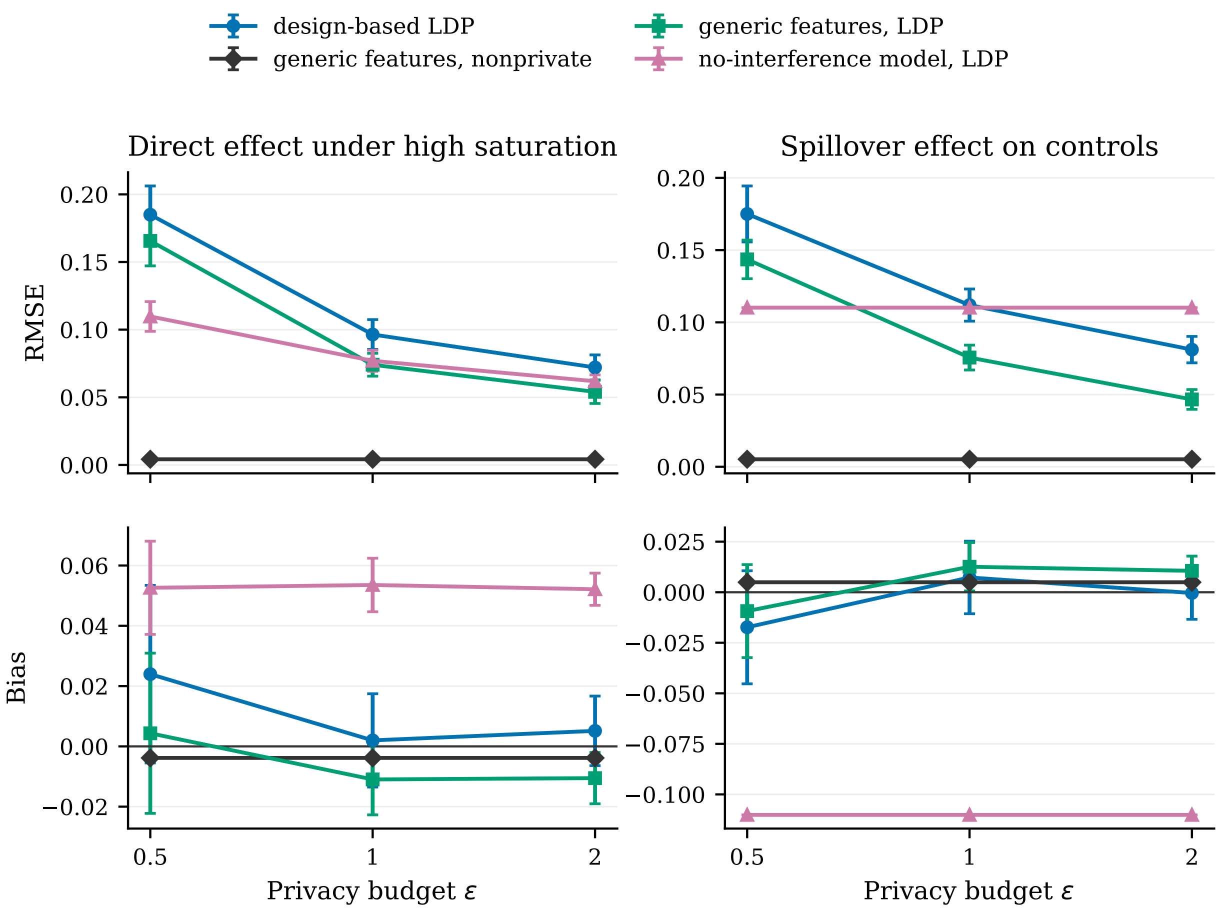

Figure 4: **Model-dependent point estimation is a bias–variance choice.** Top: RMSE; bottom: signed bias, with 95% Monte Carlo intervals. Generic polynomial network features use the exact privatized-bit likelihood and reduce RMSE here; the same nonprivate fit isolates privacy loss. Omitting interference forces the spillover estimate to zero and produces persistent bias. Only the design estimator retains a causal guarantee under either model.

9 Limitations and practical guidance

Block boundaries are an assumption. The method allows arbitrary interference only *within* known blocks. Cross-block effects invalidate the independent-block variance and can destroy the growing-replication guarantee. When approximate interference decay is plausible, network HAC or graph asymptotics may be more appropriate [Leung, 2020, 2022].

The privacy scope is outcome only. Assignments, block labels, covariates, and any network summaries are public. Jointly protecting them requires composition or a custom multivariate channel and changes the rate constants. The one-bit mechanism also assumes a declared outcome range. Clipping an unbounded outcome changes the estimand unless tail assumptions justify it.

Policy effects are not exact-exposure effects. Our spillover effect answers what happens when a block moves from p_0 to p_1. It does not answer what happens under one exact peer configuration. Theorem 8 quantifies why the latter may be far harder. Researchers should choose the effect before selecting an estimator.

Latent DoI inference remains model dependent. The private likelihood is exact, but a fitted DoI distribution is only as credible as its outcome model and public interference summaries. We report a misspecification ablation and keep the design estimator as the primary analysis. Formal Bayesian calibration for a full dependent Dirichlet process under shrinking privacy is left open.

Large numbers of blocks matter. The t interval is asymptotic in C, not in the raw user count. A trial with thousands of users in a handful of interacting villages still has weak design-based replication. Power calculations should therefore vary both C and m, using $C^{-1} + (Cma_\epsilon^2)^{-1}$ rather than N^{-1} alone.

10 Conclusion

Local privacy does not make unknown interference disappear, and a flexible latent model does not by itself identify a causal effect. A randomized saturation design changes the situation: it defines direct and spillover policy effects that average over the unknown DoI, while one private bit per outcome supports unbiased estimation and cluster-level inference. The minimax analysis separates two resources—independent blocks and independent local messages—and shows when they combine additively. Conditioning on a rare DoI cell produces a different, multiplicative privacy–overlap cost. This distinction is the practical takeaway: design for policy-relevant marginal effects when possible, report exposure probabilities when not, and use latent DoI models as transparent sensitivity analyses rather than hidden identification assumptions.

A Proofs

A.1 Proof of the privacy lemma

Let $q(y) = b_\epsilon + a_\epsilon y$. Since $q(0) = 1/(1 + e^\epsilon)$ and $q(1) = e^\epsilon/(1 + e^\epsilon)$, $q(1)/q(0) = e^\epsilon$. Also $(1 - q(0))/(1 - q(1)) = e^\epsilon$. Both $q(y)$ and $1 - q(y)$ are affine, so the largest ratio over $y, y' \in [0, 1]$

occurs at the endpoints. This proves ϵ-LDP. Moreover,

$$\mathbb{E}(U \mid Y = y) = \frac{q(y) - b_\epsilon}{a_\epsilon} = y,$$

and

$$\operatorname{Var}(U \mid Y = y) = \frac{q(y)\{1 - q(y)\}}{a_\epsilon^2} \le \frac{1}{4a_\epsilon^2}.$$

A.2 Proof of exact unbiasedness

Condition first on the realized raw outcome. By lemma 4, replacing U_{ci} by $Y_{ci}(\mathbf{Z}_c)$ does not change the expectation. For a fixed unit,

$$\mathbb{E}\left[\frac{\mathbf{1}\{S_c = s, Z_{ci} = z\}}{r_s q_{sz}} Y_{ci}(\mathbf{Z}_c)\right]$$
$$= \mathbb{E}_{\mathbf{Z}_{c,-i} \sim \text{Bernoulli}(p_s)^{m-1}}\{Y_{ci}(z, \mathbf{Z}_{c,-i})\},$$

because the inverse probabilities cancel the probabilities of $S_c = s$ and $Z_{ci} = z$. Averaging units and blocks gives (2). Linearity gives the four contrasts.

A.3 Proof of the block central limit theorem

Under assumption 1, the H_c in (7) are independent and identically distributed. Fixed positive ϵ and overlap bound every possible decoded bit and every weight, so H_c has finite second moment. The iid central limit theorem yields $\sqrt{C}(\widehat{\tau} - \tau) \Rightarrow \mathcal{N}(0, \operatorname{Var} H_c)$. The sample variance of H_c is consistent for $\operatorname{Var} H_c$, and Slutsky's theorem gives the result. Triangular arrays with shrinking ϵ or growing m require a corresponding Lindeberg condition; the fixed setting stated in the theorem avoids hiding that extra assumption.

A.4 Proof of the matched minimax theorem

Upper bound. Write $H_c = \mathbb{E}(H_c \mid \mathcal{D}_c) + \xi_c$, where $\mathcal{D}_c$ contains the complete raw block experiment and ξ_c is privacy noise. Bounded raw outcomes and overlap imply $\operatorname{Var}\{\mathbb{E}(H_c \mid \mathcal{D}_c)\} \le K_{1,\eta}$. Conditional independence of the local coins, lemma 4, and the m^{-1} block average imply

$$\mathbb{E}\{\operatorname{Var}(H_c \mid \mathcal{D}_c)\} \le \frac{K_{2,\eta}}{m a_\epsilon^2}.$$

Independence across blocks then gives

$$\mathbb{E}(\widehat{\tau} - \tau)^2 \le \frac{K_{1,\eta}}{C} + \frac{K_{2,\eta}}{C m a_\epsilon^2}.$$

Clipping the estimate to the effect range cannot increase squared error and truncates the bound at a constant.

Interference lower bound. Restrict the parameter class to bounded schedules with $V_c \overset{\text{iid}}{\sim}$ Bernoulli(θ) and genuine within-block spillover:

$$Y_{ci}(\mathbf{z}_c) = \frac{1}{4} + \frac{1}{4}V_c z_{ci} + \frac{V_c}{4(m-1)}\sum_{j\neq i} z_{cj}.$$

The direct contrasts equal $\theta/4$, and the spillover contrasts equal $(p_1 - p_0)\theta/4$. Even an oracle that reveals V_c supplies only C iid Bernoulli draws. The two-point Le Cam bound therefore gives $c_{\eta,\delta}/C$ risk for every contrast [Yu, 1997, Tsybakov, 2009]. The raw and private experiments are no more informative than this oracle experiment.

Privacy lower bound. Let $W_{ci} \overset{\text{iid}}{\sim}$ Bernoulli(θ). For a direct contrast, restrict to $Y_{ci}(\mathbf{z}_c) = 1/4 + z_{ci}W_{ci}/4$, whose target is $\theta/4$. For a spillover contrast, restrict instead to

$$Y_{ci}(\mathbf{z}_c) = \frac{1}{4} + \frac{W_{ci}}{4(m-1)}\sum_{j\neq i} z_{cj},$$

whose target is $(p_1 - p_0)\theta/4$. Conditional on public assignments, every outcome-LDP report induces an ϵ-LDP channel of W_{ci} because the outcome is a deterministic function of W_{ci} and that assignment. Enlarging this induced channel family to all ϵ-LDP channels of the $N = Cm$ variables W_{ci} can only make estimation easier. Private Bernoulli mean estimation, with target scaling bounded below by $\min\{1/4, \delta/4\}$, has risk $c_{\eta,\delta}\min\{1, (N\min\{\epsilon^2, 1\})^{-1}\}$ [Duchi et al., 2018]. Since $\tanh^2(\epsilon/2)$ and $\min\{\epsilon^2, 1\}$ differ only by universal constants, this is $c'_{\eta,\delta}\min\{1, (Na_\epsilon^2)^{-1}\}$. The maximum of the interference and privacy lower bounds is at least a constant multiple of their capped sum, completing the proof.

A.5 Proof of the exposure-overlap theorem

Condition on the public exposure indicators. Their count K is binomial with mean $N\rho$. Within the cell, restrict outcomes to Bernoulli distributions whose means differ by a local two-point perturbation. Conditional on K, the LDP contraction bound for Bernoulli mean estimation gives risk at least $c\min\{1, (Ka_\epsilon^2)^{-1}\}$. If $N\rho$ is bounded, the event $K = 0$ has constant probability and the risk is constant. Otherwise, a binomial concentration event with $K \le 2N\rho$ has probability bounded away from zero, giving $c\min\{1, (N\rho a_\epsilon^2)^{-1}\}$. For the upper bound, use

$$\widehat{\mu}_e = \frac{1}{N\rho}\sum_{i=1}^{N} \mathbf{1}\{E_i = e\}U_i.$$

Its design and privacy variance is $O((N\rho a_\epsilon^2)^{-1})$; clipping to $[0, 1]$ gives the capped rate in the small-cell regime.

B Algorithms and reproducibility details

Primary estimator. Each user evaluates (4) once and sends one bit. The analyst decodes with (5), computes the four cell scores in (6), forms contrasts, and uses (8). The unprojected contrast is reported for exact unbiasedness and interval construction; its clipped version is used for bounded-loss comparisons. No iterative fitting is involved.

DoI-motivated finite-feature regression. We minimize the negative log likelihood from (11) plus an ℓ_2 penalty of 0.05 on non-intercept coefficients using L-BFGS. The generic dictionary contains polynomial rather than true generating bases; the no-interference dictionary omits all peer summaries. G-computation uses 16 fresh synthetic assignment draws from each target policy. These draws are post-processing and do not consume privacy budget. These regressions produce model-dependent point estimates only.

Monte Carlo uncertainty. For errors e_b from B replications, RMSE is $\{B^{-1}\sum_b e_b^2\}^{1/2}$. Its reported Monte Carlo standard error applies the delta method to the sample variance of e_b^2. Coverage bars use Wilson binomial intervals. All configurations, raw results, summaries, figures, and file checksums are included in the supplementary repository.

C Additional estimand clarifications

The direct effect τ_s^{dir} changes own treatment while averaging peers under the same policy p_s. The policy-spillover effect τ_z^{sp} changes peers' assignment distribution while holding own treatment fixed. Their sum along a path defines a total policy effect, but we do not report it because the two components are more diagnostic. In general, $\tau_1^{\mathrm{dir}} \neq \tau_0^{\mathrm{dir}}$ when own treatment interacts with the DoI, and $\tau_1^{\mathrm{sp}} \neq \tau_0^{\mathrm{sp}}$ for the same reason. The nonlinear simulation was constructed to include both inequalities.

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